[NOT MEASUREMENT SENSITIVE]
FED-STD-793C
February 2, 2021
SUPERSEDING
FED-STD-793B
MAY 22, 2006

FEDERAL STANDARD
DEPOT STORAGE STANDARDS

The General Services Administration has authorized the use of this federal standard by all federal agencies.

1. SCOPE.

This standard details depot storage conditions, depot inspection instructions, and shelf-life extension criteria for mostly GSA-managed shelf-life material.

2. REFERENCED DOCUMENTS.

ANSI/ASQ Z1.4, Sampling Procedures and Tables for Inspection by Attributes. Copies may be obtained from American Society for Quality (ASQ), P.O. Box 3005, Milwaukee, WI 53201-3005or from the web at ASQ.org).

Joint Service Regulation DLAR (JP) 4155.37/AR 702-18/NAVSUPINST 4410.56B/AFJMAN 23-232 (IP)/MCO 4450.13B, Department of Defense (DoD) Shelf Life Material Quality Control Storage Standards copies may be accessed from the DOD Shelf-Life Web Site at DoDM 4140.27, Volume 1 DoD Shelf-Life Management Program and DoDM 4140.27, Volume 2 Materiel Quality Control Storage Standards. A copy of Volumes 1 and 2 may be accessed from the DOD Shelf-Life Web Site at https://www.shelflife.dla.mil/

3. GENERAL INSTRUCTIONS FOR INSPECTION.

3.1 Statistical Sampling. Sampling shall be in accordance with ANSI/ASQC Z1.4, Sampling Procedures and Tables for Inspection by Attributes, unless otherwise specified.

3.2 Shelf-Life Extension Criteria. The material shall be sampled and inspected as specified in the special inspection instructions (Section 5). Where no sampling plan or inspection instructions exist, each inspecting service or agency shall follow their internal guidance or request guidance from the responsible GSA Center listed in Section 4. The material should be considered suitable for issue, and the shelf-life extended, if the number of defects is below the rejection number for the AQL specified.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration (GSA), Federal Acquisition Service (FAS), Policy, Standards,& Engineering Branch (QSDPA), 1800 F Street NW, Washington, DC 20405.

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Material failing to meet the acceptance criteria shall be disposed of in accordance with inspecting service or agency directives. For commodities not covered by Section 5, further guidance may be obtained from the responsible GSA Center listed in Section 4.

3.3 Length of Shelf-Life Extension. The method of calculating the re-inspection date shall be determined by the directives of the inspecting service or agency. The length of extension will normally be one-half (1/2) of the original shelf-life. For GSA managed items, GSA recommends that, the length of extension is one-half (1/2) of the original shelf-life time for the first inspection, one-third (1/3) of the original shelf-life time for the second re-inspection, and the remaining shelf life time for the third re-inspection. An extension period containing a fraction of a month should be rounded up to the next whole month.

3.4 Maximum Shelf-Life Extensions. Unless otherwise specified in Section 5, the re-inspection date for GSA managed Type II (extendible) items, may not be extended beyond a date determined by adding 2 times the item's original shelf-life to the date of manufacture, cure, pack, assembly, or ship; as applicable. For example, a disinfectant in FSC 7930 with a date of manufacture (MM/YY) of 05/19 and a shelf-life code of 6 (24 months) may not have its re-inspection date extended beyond 05/23. For example, the aforementioned disinfectant is extended in three increments: until 05/22 after the first inspection, 01/23 after the second re-inspection and 05/23 after the third and final re-inspection. Other government entities shall be at liberty to reach the maximum extension date allowable in two or three increments.

3.5 Shelf-Life Extensions Label. Extended items shall receive an extension label (or notice) displaying at least the new extension date in addition to the NSN, contract or purchase order numbers, lot or batch numbers, inspection or test dates, next re-inspection or test dates, and the inspector's identification.

3.6 Inspection Results and Recommended Changes. Failure of a material to pass the shelf-life inspection may indicate inadequate storage conditions, or an inappropriate shelf-life code. Repeated shelf-life extensions may indicate that a longer shelf-life code should be assigned. Recommended changes to shelf-life codes should be forwarded to the GSA Center responsible for that stock class listed in Section 4.

4. GSA ACQUISITION CENTERS

For additional information concerning GSA managed items, contact the Commodity Acquisition Center responsible for that federal supply class (FSC) or federal supply group (FSG). Listed below are the GSA Acquisition Centers and the FSCs/FSGs of responsibility:

GSA
Office of Fleet Management
FAS-QMD
1800 F St., NW, Washington, DC 20406
703-605-3413

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FSC/FSG: 2310, 2320, 2330, 24-29, 38, and 4210 (fire trucks only).

GSA
Heartland Acquisition Center (HAC)
FAS-6QSBCB
2300 Main, 2 Pershing Square
Kansas City, MO 64108
816-926-2429
FAX 926-1371
FSC/FSG: appliances in 3510, 4110, 4120, 4140, 51-52, 7310, 7320, and 91.

GSA
Integrated Workplace Acquisition Center
100 South Independence Mall West
Philadelphia, PA 19106-2320
Phone: 215-446-4872
FSC/FSG: 71, 7220 and 7230

GSA
Office Supplies & Paper Products Center
FAS-2FYE
26 Federal Plaza
New York, NY 10278
212-264-3573
FAX 264-3574
FSC/FCG: 3540, 5820, 5825, 5835, 5836, 5965, 6130, 6645, 6650, 6675, 6680, 6850, 69, 7025, 7045, 7105 (frames), 7110, 7125, 7195, 7420 (calculators), 7430 (typewriter covers), 7460, 7490 (embossing machines), 7510, 7520, 7530, 7610, 7690, 8105, 8110, 8115, 8125, 8135, 8540, 9310.

GSA
Southwest Supply Center
FAS-7FLE
819 Taylor Street
Fort Worth, TX 76102
817-574-2390
FAX 574-2395
FSC/FCG: FORG, MECA, MFFP, MSDS, THDS, 10, 12, 13, 15, 16, 17, 19, 20, 22, 23, 2805, 30, 31, 3510 (except laundry equip.), 3540, 3550, 3590, 36 (except 3610 & 3615), 38, 39, 4010, 4020, 4030, 4130, 4210, 4220, 4230, 4235, 4240 (except fire trucks), 4250, 4310, 4220, 4330, 4410, 4420, 4430, 4440, 4460, 4470, 4510, 4520, 4530, 4540, 46, 47, 48, 4910, 4920, 4921, 4923, 4925, 4927, 4930, 4931, 4933, 4935, 4940, 4953, 4960, 51 (fire items only), 5305, 5306, 5307, 5310, 5315, 5320, 5325, 5330, 5335, 5340, 5342, 5345, 5350, 5355, 5360, 5410, 5411, 5420, 5430, 5440, 5445, 5450, 5510, 5520, 5530, 5610, 5620, 5630, 5640, 5650, 5660, 5670, 5675, 5680, 5910, 5915, 5920, 5925, 5925, 5930, 5935, 5940, 5945, 5955, 5960, 5961, 5962, 5963, 5965, 5970, 5975, 5977, 5985, 5990, 5995, 6010, 6015, 6020, 6030, 6060, 6070, 6080, 6105, 6110, 6115, 6116, 6117, 6117, 6120, 6125, 6135, 6140, 6145, 6150, 6210, 6220, 6230, 6240, 6250, 6260, 6310,

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6320, 6330, 6340, 6350, 6545, 6600, 6605, 6610, 6615, 6620, 6625, 6635, 6636, 6640, 6650, 6655, 6660, 6665, 6670, 6675, 6680, 6685, 6695, 6700, 6720, 6730, 6740, 6750, 6770, 6780, 6810, 6820, 6830, 6840, 6850,6930 (fire items only), 7310, 7320, 7330, 7340, 7350, 7360, 7530 (forms), 7540 (forms), 77, 78, 7910, 7920, 7930, 8305, 8310, 8315, 8320, 8325, 8330, 8335, 8340, 8345, 84, 8510, 8530, 8540, 8710, 8720, 88, 9110, 9130, 9135, 9140, 9410, 9420, 9430, 9440, 9450, 9515, 9520, 9530, 9540, 9545, 9610, 9620, 9630, 9640, 9650, 9905, 9910, 9920, 9925, 9930.

5. SPECIAL INSPECTION INSTRUCTIONS FOR SELECTED COMMODITIES.

5.1 FSC 4210 - Fire Fighting Equipment

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings shall be securely attached, clear and legible.

For fire-fighting chemicals, refer to paragraph 5.12, FSC 6810. The first and second re-inspections for this material shall be in accordance with the first inspection for FSC 6810. For non-metallic hose and hose assemblies, on the first re-inspection, the sample shall be removed from the storage container, fully extended, and examined visually for debris, mildew, rot and other physical damage. The inspection shall include all visible components of the hose/hose assembly. On the second re-inspection, the sample shall be evaluated in accordance with the latest edition of NFPA 1962, Standard for the Care, Use and Service Testing of Fire Hose Including Couplings and Nozzles.

5.2 FSC 4240 – Respiratory Masks with Air Lines

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

The regulator shall show no signs of corrosion. The breathing tube, upon being stretched, shall show no evidence of cracks or holes. The face mask, after being subjected to light bending and twisting, shall show no evidence of cracks or flaking. The head strap shall show no signs of corrosion to metal parts, fraying of cloth parts, or cracking of rubber parts.

5.3 FSC 4240 – Industrial Gas Masks

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

One end of the breathing tube assembly shall be subjected to a pressure controlled air source with pressure regulator set to furnish 10 inches of water with the other end of the

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tube sealed tightly. The breathing tube assembly shall then be submerged in water to test for leaks. No leaks shall be permitted. As a condition for extending the shelf life of the gas mask, the accompanying canisters shall be replaced.

5.4 FSC 5640 – Duct Tape

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

The duct tape shall be visually examined for defects which may impair usefulness. Adhesive remains active, tape does not tear when unwound. If adhesion does not appear adequate or if the tape strength appears compromised, test one roll for adhesion and one roll for tensile strength. Minimum adhesion shall be 40 ounces per inch width in accordance with ASTM D3330, Procedure A. Minimum tensile strength shall be 25 pounds per inch width in accordance with ASTM D3759.

5.5 FSC 5970 – Electrical Coating

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

The electrical coating, upon being brushed onto a test splice, shall demonstrate good fluidity, permitting an effective coating of the splice.

5.6 FSC 5970 – Insulation Tape, Electrical, Pressure Sensitive Thermosetting Adhesive

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

A short segment of tape shall be unrolled and then cut or torn from the roll. The segment of tape shall have good adhesion without any fraying or other damage.

5.7 FSC 5970 – Insulation Tape, Electrical, Friction

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

A short segment of tape shall be unrolled and then cut or torn from the roll. The segment of tape shall have good adhesion without any fraying or other damage.

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5.8 FSC 6230 – Light Sets, Emergency and Auxiliary

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

Sealed maintenance-free batteries shall not show any evidence of leakage or terminal corrosion.

5.9 FSC 6505 - Antiseptic Kit

Inspection plan: Level 1, AQL 1.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

The following components of the Antiseptic Kit shall be visually inspected:

Povidone Iodine bottle shall be inspected for evidence of Iodine leakage around the cap area. Tamper-evident seal shall be intact and show no sign of tampering. The kit shall be inspected for evidence of gas permeating through the walls of the bottle. Such evidence includes staining or discoloration of surrounding items within the kit.

Povidone Iodine swabs shall be inspected for evidence of leakage from the Iodine ampoule. The ampoule shall be intact and show no sign of damage or other contamination.

5.10 FSC 6545 - First Aid Kit, General

Inspection plan: Level 1, AQL 1.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

The following components of the First Aid Kit shall be visually inspected. Eyewash Solution container shall be inspected for evidence of leakage. Tamper-evident seal shall be intact and show no sign of tampering. Aspirin Tablet packets shall be inspected for conditions such as torn or damaged packaging, stains or discoloration, or evidence of soiling or other contamination.

Acetaminophen Tablet packets shall be inspected for conditions such as torn or damaged packaging, stains or discoloration, or evidence of soiling or other contamination.

5.11 FSC 6750 - Sodium, Sulfite, Anhydrous - Photographic

Inspection plan: Level S-2, AQL 4.0 percent defective.

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Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

The physical and functional requirements for this product shall be in accordance with ISO 418 and all associated test methods.
Upon re-inspection, determine that the powder is free flowing and insure that it does not contain any clumping of material. Sodium shall be free of any foreign materials, including water or liquid. Any of these listed defects will constitute failure and rejection of the material. Unless otherwise stated on the containers, the material is recommended to be stored at room temperatures between 50 and 80 degrees F.

5.12 FSC 6810 - Chemicals.

Inspection plan: Level S-2, AQL 4.0 percent defective.

Containers shall be secure and free from leaks, rust and other contaminants, and bulges. Markings shall be securely attached, clear and legible.

On the first re-inspection, liquid products shall be homogeneous solutions with no separation, sediment, putrefaction or other degradation. Solid products shall be free flowing with no significant water absorption or other contamination.
On the second re-inspection, the material shall be analyzed for the active ingredients specified on the label. If the level of active ingredients is within 10 percent of that specified on the label, the shelf-life may be extended once more.

Material shall be stored at temperatures between 32 and 90 degrees F (0 and 32 degrees C).

5.13 FSC 6840 - Disinfectant, Disinfectant-Detergent, and Sanitizer-Detergent. See paragraph 5.12.

5.14 FSC 6850 - Miscellaneous Chemical Specialties. See paragraph 5.12.

5.15 FSC 7350 Tube, Drinking, Plastic (STRAW)
Inspection Plan: Level S-2, AQL 4.0 percent defective
Unit, intermediate, and shipping containers shall be secure and free from leaks, contaminants, or significant damage. Markings on all containers shall be securely attached, clear, and legible.

Randomly select several boxes and examine the paper wrapping covering the drinking straws. The paper wrapping shall be free of holes, dirt, and discoloration.

5.16 FSC 7510 Inks.

The following criteria apply to the subparagraphs below.

Inspection plan: Level S-3, AQL 2.5 percent defective.

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Unit, intermediate, and shipping containers shall be secure and free from leaks, contaminants, dents, bulges, or other distortion. Markings on all containers shall be securely attached, clear, and legible.

The ink shall be in uniform suspension with no settling or caking.
5.16.1 Ink, Drawing (Waterproof, Colored)
5.16.2 Inks, Marking Stencil, Opaque
5.16.3 Inks, Stamp Pad
5.16.4 Ink, Writing Red
5.16.5 Ink, Numbering and Dating Machine
5.16.6 Ink, Writing For Fountain and Dip Pens
5.16.7 Ink, Black, For Duplicating Process
5.16.8 Ink, Stencil, Duplicating Process, Black
5.16.9 Ink, Drawing, Waterproof, Black

5.17 FSC 7510 Ribbons.

The following criteria apply to the subparagraphs below.

Inspection plan: Level S-4, AQL 2.5 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from leaks, contaminants, or significant damage. Markings on all containers shall be securely attached, clear, and legible.

Ribbons of uneven weave, with frayed edges, slits, or knots shall be cause for rejection. Performance may be evaluated by selecting one ribbon from each lot, batch or order number to test for desired characteristics. Assess the ribbon's performance in the machine for which it is specified. When used on the designated machine model, ribbons shall produce clear, clean, and legible type impressions. If performance is satisfactory, the shelf-life may be extended.

5.17.1 Ribbon, Printing, Cotton and Nylon
5.17.2 Single Strike Typewriter Ribbons
5.17.3 Ribbons, Typewriter
5.17.4 Ribbon, Teletypewriter
5.17.5 Ribbon, Accounting Machine
5.17.6 Ribbon, Machine (Word Processing)

5.18 FSC 7510 - Tape, Pressure-Sensitive

Inspection plan: Level S-2, AQL 4.0 percent defective.

Containers shall be secure and free from damage. Markings shall be securely attached, clear and legible.

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Items procured under military specifications shall be tested for, and comply with, all specification requirements. Items bought to Federal specifications or Commercial Item Descriptions (CID's) shall be tested for the following specification requirements.

Property	Test Method
Adhesion	ASTM D3330
Holding Power	ASTM D3654
Tensile Strength	ASTM D3759
Bursting Strength	ASTM D3662 (When specified and required)
Unwind Force	ASTM D3811 (When specified and required)

Pressure-sensitive adhesive tapes deteriorate more rapidly as the storage temperature increases. These items should be issued strictly by the earliest date of manufacture. Where storage temperatures exceed 80 degrees F (27 degrees C), stock levels should be decreased and stock turn increased to minimize the storage time at elevated temperatures.

5.19 FSC 7520 Pens and Markers. The following criteria apply to the subparagraphs below:

Inspection plan: Level S-4, AQL 2.5 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from leaks, contaminants, or significant damage. Markings on all containers shall be securely attached, clear, and legible.

Performance - select three pens/markers/refills at random from each lot, batch or order. Test the samples on a writing pad or paper under normal working conditions and temperature. Defects such as blobs, skips, dotting and density variations of line intensity in any of the samples shall constitute failure.

5.19.1 Pen, Roller-ball
5.19.2 Marker, Tube Type, Fine Tip, Transparency
5.19.3 Marker, Tube Type, Felt Tip
5.19.4 Marker, Tube Type, Fine, Felt Tip
5.19.5 Marker, Tube & Flat Type, Highlighter
5.19.6 Pen, Rubberized Barrel
5.19.7 Pen, Rubberized Barrel, Refills
5.19.8 Pen, Ballpoint, Non-Refillable
5.19.9 Pen, Ballpoint (Recycled)
5.19.10 Pen, Ballpoint, Refillable
5.19.11 Marker, Tube Type, Permanent
5.19.12 Marker, Assortment, Tube Type
5.19.13 Marker, Tube Type, White
5.19.14 Pen, Ballpoint, Cushion Grip

5.20 FSC 7520 Letter Set, Polystyrene.

Inspection Plan - Level S-2, AQL 4.0 percent defective.

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Unit, intermediate, and shipping containers shall be secure and free from leaks, contaminants, or significant damage. Markings on all containers shall be securely attached, clear, and legible.

Select one set from each lot, batch or order to verify desired performance. Randomly select several letters from the set and affix them to the plastic strips in accordance with the manufacturer's instructions. If the adhesion is satisfactory on test sample after eight hours, the shelf life may be extended.
Pressure-sensitive adhesive backed letters deteriorate more rapidly as the storage temperature increases. These types of items should be issued strictly by the earliest date of manufacture. Where storage temperatures exceed 80 degrees F (27 degrees C), stock levels should be decreased and stock turn increased to minimize the storage time at elevated temperatures.

5.21 FSC 7930 - Cleaning and Polishing Supplies.

Inspection Plan - Level S-2, AQL 4.0 percent defective.

Containers shall be secure and free from leaks, rust and other contaminants, and bulges. Markings shall be securely attached, clear and legible.

Liquid products shall be homogenous solutions with no separation, sediment, putrefaction or other degradation.

Solid products shall be free flowing with no significant water absorption or other contamination.

Material shall be stored at temperatures between 40 and 80 degrees F (4 and 27 degrees C).

5.22 FSC 8520 – Hand Cleaner and Skin Cleanser, Liquid

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

Liquid soap, upon visual inspection, shall not show any evidence of separation of the ingredients. Liquid soap shall not exhibit a rancid odor.

5.23 FSC 8520 – Shampoo and Body Wash

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

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Shampoo and body wash, upon visual inspection, shall not show any evidence of separation of the ingredients. Shampoo and body wash shall not exhibit a rancid odor.

5.24 FSC 8520 – Towelettes, Skin Cleansing, Individually Packaged

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

Towelettes shall not show any signs of leakage. A towelette shall be opened, and shall have sufficient moisture to enable cleaning of hands.

5.25 FSC 8520 – Bar Soap and Milled Cake Soap

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

Bar soap and milled cake soap shall be visually inspected and shall not show any discoloration. Bar soap shall not exhibit a rancid odor.

5.26 FSC 8520 – Hydrated Sodium Borate

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

Hydrate sodium borate, upon visual inspection, shall not be congealed, but shall remain in a powdered state.

5.27 FSC 9330 - Plastic Sheet

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

The adhesive layer shall not have separated from the plastic sheet. Test for minimum spectral transmittance in accordance with ASTM D 1003. Test for minimum tensile strength at yield in accordance with ASTM D 638 or D 882.

5.28 FSC 9390 – Reflective Tape

Inspection plan: Level S-2, AQL 4.0 percent defective.

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Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

The adhesive layer shall not have separated from the reflective tape. Test for minimum coefficient of retro- reflection in accordance with ASTM D 4956.

5.29 FSC 9905 Holder, Card-Label – Adhesives

Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

The minimum peel or stripping of adhesive shall be in accordance with ASTM D903. No part of the surface to which the adhesive is applied shall have bare spots or air pockets. The adhesive shall be of uniform thickness and there shall be no tendency, when removing the protective backing, for the adhesive to separate from the plastic tubing. Holders shall show no evidence of deterioration or change in properties of the adhesive or protective backing such as to make the holders unfit for use when stored under warehouse conditions at a temperature of 70 to 90 degrees F. and 50 percent plus or minus 2 percent relative humidity for one year.

5.30 FSC 9905 Ribbon, Surveyor's Flagging
Inspection plan: Level S-2, AQL 4.0 percent defective.

Unit, intermediate, and shipping containers shall be secure and free from damage. Markings on all containers shall be securely attached, clear, and legible.

The ribbon shall be free of holes, dirt, ragged edges, and discoloration. No defects that affect appearance or serviceability shall be permitted. Tensile strength and elongation shall be determined in accordance with ASTM D638. Tensile Strength – 7.0 lb, minimum. Elongation – 200 percent minimum.

6. KEY WORDS:
depot shelf storage
life shelf-life

Military Coordinating Activity:
SO

Preparing Activity:
GSA-FAS

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Appendix A

Characteristics Codes and Definitions

Characteristics codes (formerly: defect codes) are the item characteristics that are looked at when performing visual inspection on items. The listing below of codes and their meanings represent only the codes that apply to GSA managed items. These codes are used in the Material Quality Control Storage Standards (MQCSS) database to provide for the inspection criteria for type II (extendible) shelf life items. A complete listing of codes for GSA and non-GSA managed items are found in DoDM 4140.27, Volume 2 DoD Shelf-Life Management Program: Materiel Quality Control Storage Standards and Joint Service Regulation DLAR (JP) 4155.37/AR 702-18/NAVSUPINST 4410.56B/AFJMAN 23-232 (IP)/MCO 4450.13B, Department of Defense (DoD) Shelf Life Material Quality Control Storage Standards . Storage activities use the codes when adding/updating the on-line storage standard data of extendible (type II) shelf life items in the Material Quality Control Storage Standards (MQCSS) database.

DEFECT CODE	LAB VISUAL DEFECT CODE	DESCRIPTION
Q9		Reserved for future use
R2		Unsecured, leaking, rusted, contaminated, bulged, dented, distorted, or significantly damaged (containers).
R3		Not securely attached, unclear, illegible, or incorrect (markings).
	R4L	Hard settling, caking, jelling, skinning, curdling, seeding, rusting, putrefaction (the process of decay or rotting in a body
		or other organic matter), separation (Coatings).
	R5L	Not in uniform suspension with settling or caking (Ink).
	R6L	Not mixable, not able to spray or deposit uniform film, or actuator not functional (Paint Aerosol Dispensers).
	R7L	Uneven weave, with frayed edges, slits, or knots (Ribbons).
	R8L	Performance unsatisfactory (Ribbons produce unclear, unclean, and illegible type impressions).
	R9L	Separating, layering, settling, lumping, dispersing into inhomogeneous mixture when mixed (Sealants).
	S1L	Stiffness/Dryness (Leather).
	S2L	Blobbing (a soft mass or drop, as of some viscous liquid), skipping, dotting and varying density of line intensity
		(Pens/Markers/Refills).
	S3L	Not adhering satisfactorily (Letters).
	S4L	Inhomogeneous solutions with evidence of separation, sediment, putrefaction (the process of decay or rotting in a body
		or other organic matter) or other degradation (Liquid Chemicals).

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	S5L	Not free flowing with significant water absorption or other contamination (Solid Chemicals).
	S6L	Not free from grit, seeds, lumps (Coatings).
	S7L	Abnormal thickening, livering (the process of liquid thickening and becoming lumpy), pigment flotation or excessive
		settling (Coatings), traces of particulate matter.
	S8L	Trace of particulate matter suspended or settled (Coatings).
	S9L	Mixing not accomplished by specified time (Coatings).
	U3L	Evidence of reaction between the container and its contents.
U4		Container not securely sealed and contents show evidence of deterioration.
U5		Container not tightly sealed and contents show evidence of evaporation.
	U6L	Visually observable deterioration.
	U7L	Typing or printing illegible or not uniform in color on all paper copies or plies.
	U8L	Carbon coating appears cracked or flaky (paper).
	U9L	Unsmooth, inhomogeneous mixture not free of lumps, separation or crystallization.
V1		Evidence of debris, mildew, excessive moisture or dampness, rot and other physical damage (only outside packaging).
	V2L	Failed testing or evaluation; or not in accordance with standard and test methods.
V3		Evidence of materiel degradation due to weather or environment, cracks, holes, flaking, corrosion, or fraying
		(only on outside packaging).
	V4L	Evidence of leaks after testing
	V5L	Not demonstrating good fluidity.
	V6L	Not adhering well; fraying, other damages.
	W1L	Reinforcement failure - (e.g., metal straps, wire, tape.)
	W2L	Skids, runners, or materiel handling aids damaged, inadequate, or deteriorated.
	W3L	Blocking and/or bracing inadequate.
	W4L	Dark brown or black (not white) liquid floor wax.
	W5L	Rancid (smelling or testing unpleasant because of being old or stale) - waterless hand cleaner.
	W6L	Separation into components
	W7L	Brittle and water absorption unsatisfactory (paper towels).
	W8L	Adhesive layer has separated from the plastic sheet.
	W9L	Ragged edges, cracks, pits, and dirt. Surface not smooth and not flat. Backing not completely covered by
		protective liner (Reflective sheeting)
	X1L	Evidence of leakage or battery terminal corrosion.
	X2L	Evidence of Iodine leakage; gas permeating through bottle wall.

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	X3L	Kit components not intact, damaged, contaminated, tampered with; or stained or discolored kit surroundings.
	X4L	Container leaking; seal not intact or tampered with; tablet packets torn, damaged, stained, discolored, soiling,
		or other contamination.
	X5L	Powder not free flowing without clumping or not free of foreign materials.
	Y1L	Straw Paper wrapping not free of holes, dirt, and discoloration.
	Y2L	Evidence of ingredient separation.
	Y3L	Rancid (smelling or testing unpleasant because of being old or stale) odor.
	Y4L	Signs of leakage or not having sufficient moisture (towelettes).
	Y5L	Showing discoloration.
	Y6L	Congealed (having become semisolid, especially on cooling) or not in powdered state.

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